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REINFORCEMENT LEARNING
Learning with Option
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So how do you learn with this how do you learn that now that you know how to define options can I solve problems by using these options this is where SMDP Q learning comes in, all of you remember SMDP Q learning, right so I am going to update I mean the normal, if you take an action that part is clear I mean that is a normal Q learning, right in a particular state suppose I take a, option instead.

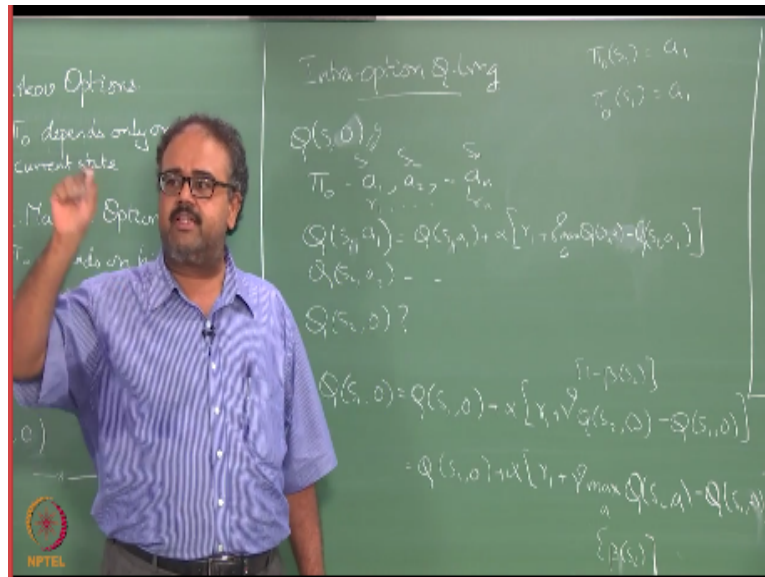
So what do I do, I treat this as a durative action, right in the SMDP Q-learning what happened we assume that actions have duration right so here in hierarchical RL, I start off with an MDP, right I have defined a set of options on top of that MDP, right so when I use these options on that MDP I am going to treat these as durative actions, so the MDP plus options gives me a SMDP, good question you could right I will come to that in a minute right.

But the simplest version is you just do so you take option O in state s then you update the value of that option in the state s, no. I said I will come to that, hold on okay, right do I need to write down the SMDP Q learning update again, you guys all know it from last class right great so the only tricky thing here is the reward and the reward for the action is going to be the the sum of the the return that you get, from when the option started till when the option ended, right.

So that is basically it. So last time we had started from r_t and ended at $r_{t+\tau}$ and then we gave appropriate discounts and everything so all of that will come into play here because, because the option is the durative action, right so that is essentially why we introduced the SMDP framework

in the first place so that we could handle all of these things, great so SMDP Q learning is fine, right but then we want to do something better right.

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So we want to do, what is called intra option Q-learning right, so we actually want to take advantage of the fact that the option consists of, primitive actions, right and we want to go and, use that as well right so what would we do in that case, is I will update $Q(s, o)$ in the normal way, right let us assume that when I start π_o right so I do actions, these are the actions I see okay this is this is because of the definition of π_o and the sequence of states S_1, S_2, S_3 that I am seeing, right

So I see let's say that I see S_1, S_2, S_n and after I do a_n and S_n I terminate, okay let us assume that this is the sequence of things I have seen okay, so what I can do is I can actually update $Q(s, a_1)$ right so how will I update $Q(s, a_1)$. what goes in here max, okay $\max_a Q(S_2, a)$.

$$Q(S_1, a_1) = Q(S_1, a_1) + \alpha [r_1 + \gamma \max_a Q(S_2, a) - Q(S_1, a_1)]$$

I cannot use a_2 in that expression because, why because it is Q learning why wouldn't I want to use a_2 in that expression because, this is where we come to your point I am following policy π_o here right.

So this is not necessarily reflective of the value of taking a_1 in state S_1 , right if I had used a_2 here it is the value of taking a_1 in S_1 assuming I am going to follow π_o thereafter and then I will after π_o terminates I will do whatever I want, right so if I had used a_2 there that is what I will be learning does that make sense normally what you expect in Q learning or sarsa is that this expression, is the reward you get for following, policy π thereafter right or following optimal policy thereafter whatever the following policy π thereafter is what you expect in sarsa.

So if I had used a_2 here I am not following policy π thereafter I am actually following π_o , right it could very well be that π_o is the policy that I would have picked there in a_2 also in S_2 also but I do not know that for sure, because I never had a choice of picking an action in S_2 I picked an action in S_1 I was constrained to follow that action in S_2 also since I didn't have the choice of picking the action in S_2 I really should not use a_2 there and claim that it is sarsa okay.

So that is why in all this kind of intra option, kind of learning mechanisms you have to use Q learning you have to be off policy, okay is there anything else I can do other than update so likewise I can do it for a_2 and so on right I can update Q, is there anything else I can do can I update any other value function, are we maintaining.. well it is like an MDP that has 10 different actions that you can take right, 4 were primitive actions 6 were options, so I am maintaining a Q function for the 4 primitive actions and the 6 options also, it is just a single Q function and it is for solving the original MDP problem, essentially I am optimizing the reward function given in that MDP. No no no I am not solving .. sub problems have all been solved.. they have been encapsulated and given to me as an option policy. I am not changing π_o anywhere.

yeah, because I am allowed to take the primitive actions also, right when I come to the state S_1 next time I can either choose to take O or I can choose to take a_1 or a_2 or a_3 or whatever right I am allowed to take primitive actions, change π_o .

So I will talk about that when I come to max Q now in the options framework itself as it was originally proposed they never had, a option of changing the option policy, right so they just let you they just assume that the option policy was frozen, and then they said okay here go learn right they never did this multi-level learning it is possible to think of mechanisms for doing that as well right then the max Q framework made it explicit, right.

So there are ways of learning the policy in the options framework also I mean it is nothing very deep or anything you just have to be very careful about how you do the book keeping so you have to make sure that the appropriate reward function is fed into the appropriate value function so I will explain that more when we talk about Max Q, which is one of the other architectures basically, so anything else that we can update, how about that option gets, option is ended, can I update this when the option is ended, option is..for what this is $Q(S_2, o)$ can I write an expression like this for $Q(S_2, o)$ is my question, when do we have an option of taking o in S_2 , great anything else the option has to be Markov right.

So what $Q(S_2, o)$ would be if o starts from S_2 and runs what will be the value, but o has started in S_1 right if it is, if it is semi Markov option then the policy would be different because I started in S_1 when I call o here, right so I should not be using this trajectory for updating the value of S_2 , o right but if o happens to be a Markov option it does not matter whether I start in S_1 , whether I start in S_2 or I start in S_3 . I can start in S_2 and I can update the value of $Q(S_2, o)$ right because it is likely to give me the same sequence of actions I mean it is a valid sequence of actions right but the only thing is the SMDP reward that I am using should start start summing from r_2 , not r_1 .

So I should leave out r_1 I should start summing from r_2 okay, great so assume I am with Markov option is there something more that I can do, assuming I am doing Markov options is there a more efficient way of updating the value of $Q(S, o)$. very good point.. and $\gamma S_2 a_2$ is this fine, does that sound right.. ahh tell me why.

$$Q(S_1, o) = Q(S_1, o) + \alpha[r_1 + \gamma Q(S_2, o) - Q(S_1, o)]$$

We do not know if π_o is the max right we do not know π_o has been defined it is in the policy that has been created for me right.

I don't know if I do Max whether I get π_o or not, on the other hand I am constrained to take actions according to π_o , no I have chosen o in S_1 right so in S_2 I have to take actions according to π_o right π_o not π_o right so I have to take actions according to π_o , I am constrained that way except when when I can end at S_2 . When I can end at S_2 , if I end at S_2 then I have to do a Max right.

So now I do not have to keep a running sum of all the returns right I can get by doing this kind of a bootstrapping, so I am writing it separately so that it is clear but you can actually do this combination of these two into a single update by using your β , right so with probability $1 - \beta$, this is what you will do and with probability β this is what you will do, so if you are terminating $\beta(S_2)$ so if you are terminating in S_2 right so then $\beta(S_2)$ will be one right.

$$Q(S_1, o) = Q(S_1, o) + \alpha[r_1 + \gamma Q(S_2, o) - Q(S_1, o)] \text{ if o not ending at } S_2$$

$$Q(S_1, o) = Q(S_1, o) + \alpha[r_1 + \gamma \max_a Q(S_2, a) - Q(S_1, o)] \text{ if o terminates at } S_2$$

Combining the above two equations into a single equation,

$$Q(S_1, o) = Q(S_1, o) + \alpha[r_1 + \gamma Q(S_2, o) - Q(S_1, o)] \times [1 - \beta(S_2)] + \alpha[r_1 + \gamma \max_a Q(S_2, a) - Q(S_1, o)] \times \beta(S_2)$$

So this will not happen so you will multiply this by $1 - \beta(S_2)$ here so if you are terminating in S_2 then this will not happen if you are not terminating in S_2 then this will happen basically for β

will be 0 and this will happen so essentially I can multiply this by, and multiply this by and then I can write everything combined into a single function, make sense everyone on board so we have done all kinds of complex things, so the easiest thing of course makes your life a lot easier to implement is SMDP Q learning but then you can do intra option Q-learning you can do a whole bunch of things right in fact you can do even something better what else can you do what else you think you can do now that you know how to do this, suppose my.

So my $\pi_o(S_1)$ is a_1 and $\pi_{o'}(S_1)$ is also a_1 some.. these are a_1 not not the first action but the the first index action not the first action according to time but the first action according to some index right essentially what I am saying is both of them give me the same action for S_1 , right if I am in state S_1 both π_o and $\pi_{o'}$ give me the same action now can you think of how I can update given that I have taken action option o I am executing option o, can I update $Q(S, o)$, what is that..no function approximation here I can do the same thing exactly so I can do the same thing right.

So I am just assuming that you can multiply those things right I will just write the first expression alone this is amazing thing right so as far as the update equations are concerned I am only looking at $S_1 r_1$, right I do not care whether that $S_1 a_1 r_1$ basically that is the combination I am looking at I do not care whether the a_1 came from executing π_o or $\pi_{o'}$, so I can still do this updation.

So if you do a single option execution I can actually update the value function for all the options which are consistent with that option that I am executing right, now let us ask me let us ask another question. Oh $Q(S_2, o)$ I already said right I can you can do that here yeah yeah same kind of bootstrapping, this is exactly where the whole off policy ratio updating thing got introduced into the RL literature, not not in chapter 5 right.

So when Doina Precup and Rich Sutton were working on this intra option Q learning then they came up with this insight that hey even if the options, I mean they first said that if the options are

consistent, if they agree with each other then I can use the same experience, and update the values of different options and then they came up with this insight, hey even if they do not agree exactly with each other as long as they are consistent in the important sampling sense, right I can use the important sampling ratio and update the values right.

So that is the insight that they came up with and then they went back and say hey why did I have to be in the options space, right I just need two different policies the behavior policy and the estimation policy right this is what happens right your behavior policy is π_o , your estimation policy is π_e . sorry. π_e is your estimation policy, your behavior policy is π_o , then they said okay this is a great thing we see we can do more generalization we can learn so we can actually maintain this huge library of options right that differ from one another with in some small respects but I can learn about all of those options, while I am executing only one option.

So that is the insight that led to this whole off policy set up that we looked at in the Monte Carlo chapter, and then they went back and then they started talking about off policy learning and in fact this is where the on policy off policy, the dichotomy that we talked about, actually got came into the literature from this intra option Q learning ideas and then we went and back fitted everything we started calling Q learning as off policy sarsa as on policy and all of those but they started only in the late 90s so cool.

$$Q(S_1, o') = Q(S_1, o') + \alpha[r_1 + \gamma Q(S_2, o') - Q(S_1, o')] \times [1 - \beta(S_2)] + \alpha[r_1 + \gamma \max_a Q(S_2, a) - Q(S_1, o')]$$

So any questions on this so all of you can implement very efficient option learning algorithms right what is that, you have to come up with options first that is a great question, yeah we have to come up with the options first, still a very active open area of research, so I do not know if time permits may be next class or something or maybe in the class after the exam I will talk about attempts people have made for option discovery, and so on so forth right not too many out there but there are some interesting attempts that have been done right.

So we will talk about that in the maybe in the last class right now I want to move on talk about the other hierarchical frameworks so so far so good right so options are good yeah and I should

say that because of the simplicity of the entire option framework right options are the most popularly used, hierarchical RL things so that is why I spent so much time I went so slowly on this one thing which you should point out there are a few things which are a little dissatisfying right.

So for example the policy π_o it is not entirely clear where π_o is defined you know. so if you think about it we have the initiation set these are states where policy can start right but once it starts here can the policy go everywhere, not really right it depends on the termination set if the dynamics of the problem is such that the termination set cuts you off from the initiation set right so those states you cannot you cannot actually reach, right.

So there is no there is no clear specification of you know admissible states for the option these are the states that this option will visit and things like these are all left to be inferred from other things that have been defined, right so what else is missing is things like whatever I was talking to you about here right so when I am actually executing option o when I come to a state I will behave differently even if it is a Markov option right I will behave differently, if as opposed to when I am executing option o' , right but for an outside observer it will look like either it will look like I am behaving in a non Markov fashion or I am behaving in a stochastic fashion right I come there I do I do action a_1 sometimes I do action a_2 sometimes it is the same state s I do action a_1 sometimes I am doing action a_2 sometimes but I might be executing a deterministic policy it depends on what option I am executing when I come to a_1 right.

The fact that the option is really part of the option you are executing is really part of the internal state is never made explicit in the options framework there is no notion of internal state or anything they are all left to you know kind of the implementers interpretation of options ok so there lot of things which the options framework really does not specify in some sense that is the power of the options framework you can go fill in whatever details you want for it but is also in some sense dissatisfying about the framework in that it lets you do a lot of this work yourself.

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